

School of Computer Science and Engineering

Winter Semester 2023-24

Continuous Assessment Test – I

SLOT E1+TE1

Programme Name & Branch: B.Tech & Computer Science and Engineering

Course Name & Code: Cryptography and Network Security & BCSE309L

Class Number (s): Applicable to all

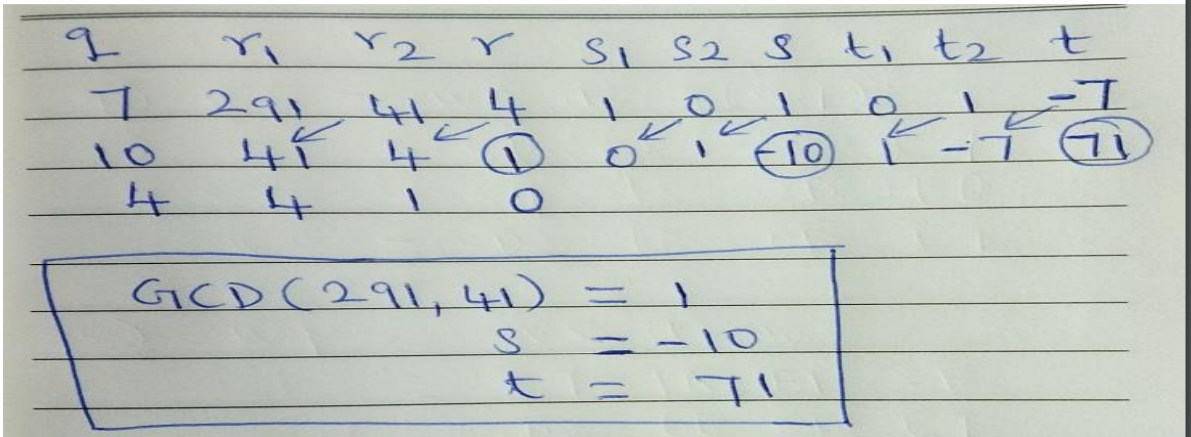
Faculty Name (s): Applicable to all

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer All the Questions and calculator is allowed

Q. No.	Question	Max Marks
1.	a) Find GCD, variables S and T by construct a table for the following inputs using Extended Euclidean Algorithm. 291, 41.  b) Find the remainder for $2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60} \bmod 7$ using Fermat's Little Theorem. Find $2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60} \bmod 7$. [Solution: $2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60} \equiv 0 \bmod 7$] By Fermat's Little Theorem, $2^6 \equiv 3^6 \equiv 4^6 \equiv 5^6 \equiv 6^6 \equiv 1 \bmod 7$. Thus, $2^{20} + 3^{30} + 4^{40} + 5^{50} + 6^{60} \equiv 2^2 + 3^0 + 4^4 + 5^2 + 6^0 \equiv 4 + 1 + 2^8 + 25 + 1 \equiv 4 + 1 + 4 + 4 + 1 \equiv 14 \equiv 0 \bmod 7$.	5 5

2. A hoard of gold pieces comes into the possession of a band of 15 pirates. When they come to divide up the coins, they find that 3 are left over. Their discussion of what to do with these extra coins becomes animated, and by the time some semblance of order returns there remain only 7 pirates capable of making an effective claim on the hoard. When, however, the hoard is divided between these seven it is found that 2 pieces are left over. There ensues an unfortunate repetition of the earlier disagreement, but this does at least have the consequence that the 4 pirates who remain are able to divide up the hoard evenly between them. What is the minimum number of gold pieces which could have been in the hoard?

$$\begin{aligned}
 x &\equiv 3 \pmod{15} \\
 x &\equiv 2 \pmod{7} \\
 x &\equiv 0 \pmod{4}
 \end{aligned}$$

$\text{GCD}(15, 7)$
 $\text{GCD}(15, 4)$
 $\text{GCD}(7, 4)$
 $= 1$

1) $M = m_1 \times m_2 \times m_3$
 $M = 15 \times 7 \times 4$
 $M = 420$

2) $M_1 = M / m_1$
 $= 420 / 15 = 28$
 $M_2 = 420 / 7 = 60$
 $M_3 = 420 / 4 = 105$

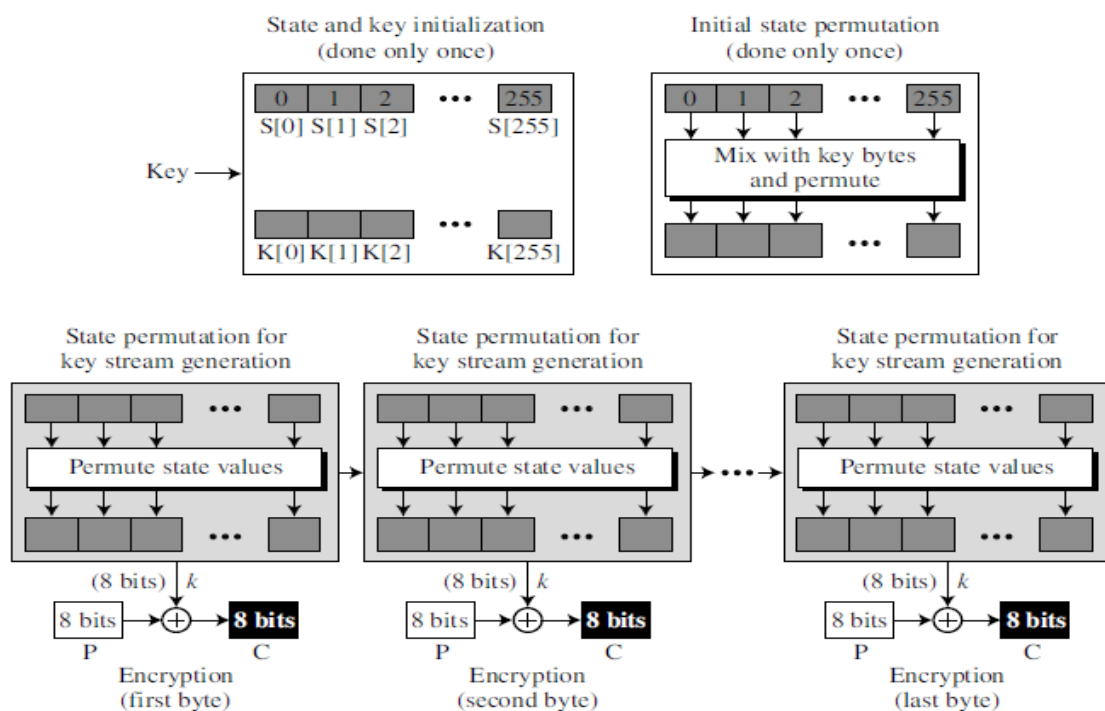
$m_1^{-1} = 28^{-1} \pmod{15}$
 $= 28 \times ? \pmod{15} = 1$
 $= 28 \times 7 \pmod{15} = 1$
 $= 196 \pmod{15} = 1$
 $m_1^{-1} = 7$

$m_2^{-1} = 60^{-1} \pmod{7}$
 $= 60 \times ? \pmod{7} = 1$
 $= 60 \times 2 \pmod{7} = 1$
 $= 120 \pmod{7} = 1$
 $m_2^{-1} = 2$

$$\begin{aligned}
 x &= [(3 \times 28 \times 7) + (2 \times 60 \times 2)] \text{ mod } 420 \\
 x &= [588 + 240] \text{ mod } 420 \\
 x &= 828 \text{ mod } 420 \\
 \boxed{x &= 408}
 \end{aligned}$$

3. Draw an architecture of RC4 algorithm and discuss the process of initialization, initial state permutation, key stream generation and encryption in detail.

10



Architecture – 2

Initialization process - 2

Initial state permutation - 2

Key stream generation - 2

Encryption - 2

4. a) Explain the DES feistel structure in detail with neat diagram.

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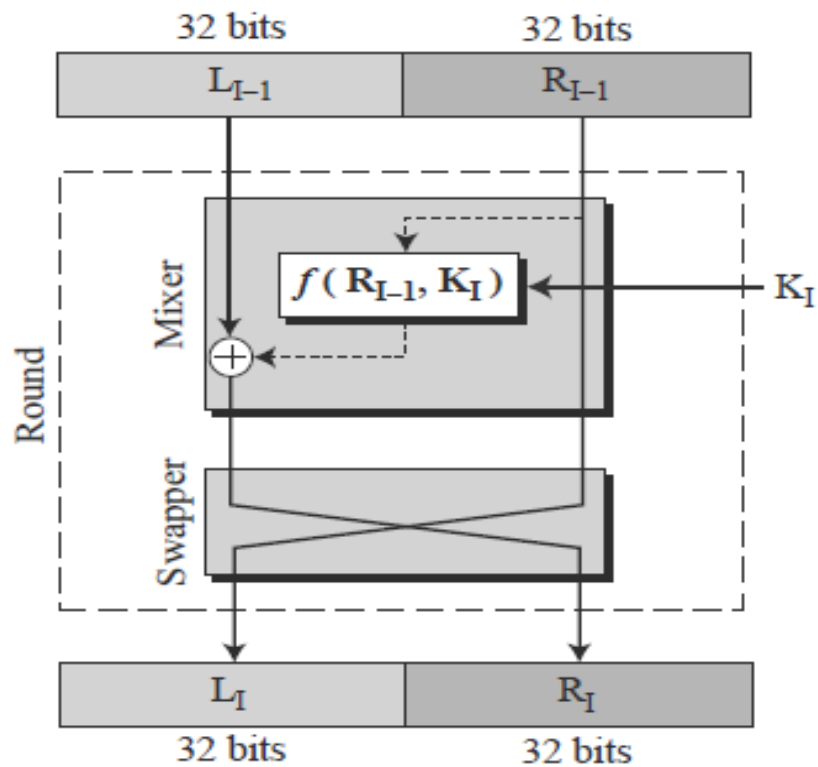


Fig. 6.4 A round in DES (encryption site)

b) Answer the following questions about S-boxes in DES:

- Show the result of passing the input 111111 through S-box 2.
- Show the result of passing the input 000000 through S-box 7.

S-box 2 Table

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	15	01	08	14	06	11	03	04	09	07	02	13	12	00	05	10
1	03	13	04	07	15	02	08	14	12	00	01	10	06	09	11	05
2	00	14	07	11	10	04	13	01	05	08	12	06	09	03	02	15
3	13	08	10	01	03	15	04	02	11	06	07	12	00	05	14	09

S-box 7 Table

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	4	11	2	14	15	00	08	13	03	12	09	07	05	10	06	01
1	13	00	11	07	04	09	01	10	14	03	05	12	02	15	08	06
2	01	04	11	13	12	03	07	14	10	15	06	08	00	05	09	02
3	06	11	13	08	01	04	10	07	09	05	00	15	14	02	03	12

i)

Input: 1 1111 1 → 3, 15 → Output: 09 (1001)

ii)

Input: 0 0000 0 → 0, 0 → Output: 04 (0100)

5. Find the third round key of AES 128 using the following second round key which is given in hexadecimal, S-Box table and round constant 04.

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Second Round Key

56	C7	76	A0
08	1A	43	3A
20	B1	55	F7
07	8F	69	FA

S-Box Table

		Y															
		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
X	0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
	1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
	2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
	3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
	4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
	5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
	6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
	7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
	8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
	9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
	A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
	B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
	C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4D	BD	8B	8A
	D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
	E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
	F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

Second Round Key = 56 C7 76 A0
 08 1A 43 3A
 20 B1 55 F7
 07 8F 69 FA
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 W4 W5 W6 W7

Find

$$g(W7) = g(A0 \ 3A \ F7 \ FA)$$

1) Rotword one byte circular shift
 = 3A F7 FA A0

2) Sub word = 80 68 2D E0

3) EX OR with RC
 = 80 68 2D E0 XOR
 04 00 00 00

$$g(W7) = 84 68 2D E0$$

$$4) W_8 = W_4 \oplus g(W_7)$$

$$= 56 \ 08 \ 20 \ 07$$

$$84 \ 68 \ 2D \ E0$$

$$W_8 = D2 \ 60 \ 0D \ E7$$

$$5) W_9 = W_5 \oplus W_8$$

$$= C7 \ 1A \ B1 \ 8F$$

$$D2 \ 60 \ 0D \ E7$$

$$W_9 = 15 \ 7A \ BC \ 68$$

$$6) W_{10} = W_6 \oplus W_9$$

$$= 76 \ 43 \ 55 \ 69$$

$$15 \ 7A \ BC \ 68$$

$$W_{10} = 63 \ 39 \ E9 \ 01$$

$$\oplus W_{11} = W_7 \oplus W_{10}$$

$$= A0 \ 3A \ F7 \ FA$$

$$W_{11} = 63 \ 39 \ E9 \ 01$$

$$W_{11} = C3 \ 03 \ 1E \ FB$$

Round 3 key:

D2 60 0D E7 15 7A BC 68 63 39 E9 01 C3 03 1E FB



School of Computer Science and Engineering

Winter Semester 2023-24

Continuous Assessment Test – I

SLOT: E2 + TE2

Programme Name & Branch: B. Tech CSE

Course Name & Code: Cryptography and Network Security – BCSE309L

Class Number (s): Applicable to All

Faculty Name (s): Prof. S. Ramani

Exam Duration: 90 Min.

Maximum Marks: 50

General instruction(s):

Answer all the Questions.

Tables related to DES/AES are given in the question paper.

Q. No.	Question	Max Marks
1.	a) Find the greatest common divisor d of 412 and 54, and find integers x and y solving the equation $412x + 54y = d$. Ans: $d = 2$. The extended Euclidean algorithm gives $x = 8$ and $y = -61$. (There are other solutions for x and y ; these are not unique.) b) Solve the congruence $X^{103} \equiv 4 \pmod{11}$ to find the value of X using Fermat's theorem. Ans: By Fermat's Little Theorem, $x^{10} \equiv 1 \pmod{11}$. Thus, $x^{103} \equiv x^3 \pmod{11}$. So, we only need to solve $x^3 \equiv 4 \pmod{11}$. If we try all the values from $x = 1$ through $x = 10$, we find that $5^3 \equiv 4 \pmod{11}$. Thus, $x \equiv 5 \pmod{11}$.	5 5
2.	a) Solve the given congruence equations to obtain the value of x $x \equiv 2 \pmod{5}$ $x \equiv 3 \pmod{7}$ $x \equiv 10 \pmod{11}$ Ans: $x \equiv 87 \pmod{385}$ b) Find $\phi(1200)$ Ans. 320	8 2
3.	a) Solve $7^{106} \pmod{143}$ using fast modular exponentiation	5 5

	106 = 1101010 in binary $106 = 2^1 + 2^3 + 2^5 + 2^6$ $106 = 2 + 8 + 32 + 64$ $7^{106} \bmod 143 = 7^{(2+8+32+64)} \bmod 143$ $7^{106} \bmod 143 = (7^2 \cdot 7^8 \cdot 7^{32} \cdot 7^{64}) \bmod 143$ Step 2: Calculate mod C of the powers of two <= B $7^1 \bmod 143 = 7 \bmod 143 = 7$ $7^2 \bmod 143 = (7^1 \cdot 7^1) \bmod 143 = (7^1 \bmod 143 \cdot 7^1 \bmod 143) \bmod 143 = (7 \cdot 7) \bmod 143 = 49 \bmod 143 = 49$ $7^4 \bmod 143 = (7^2 \cdot 7^2) \bmod 143 = (7^2 \bmod 143 \cdot 7^2 \bmod 143) \bmod 143 = (49 \cdot 49) \bmod 143 = 2401 \bmod 143 = 113$ $7^8 \bmod 143 = (7^4 \cdot 7^4) \bmod 143 = (7^4 \bmod 143 \cdot 7^4 \bmod 143) \bmod 143 = (113 \cdot 113) \bmod 143 = 12769 \bmod 143 = 42$ $7^{16} \bmod 143 = (7^8 \cdot 7^8) \bmod 143 = (7^8 \bmod 143 \cdot 7^8 \bmod 143) \bmod 143 = (42 \cdot 42) \bmod 143 = 1764 \bmod 143 = 48$ $7^{32} \bmod 143 = (7^{16} \cdot 7^{16}) \bmod 143 = (7^{16} \bmod 143 \cdot 7^{16} \bmod 143) \bmod 143 = (48 \cdot 48) \bmod 143 = 2304 \bmod 143 = 16$ $7^{64} \bmod 143 = (7^{32} \cdot 7^{32}) \bmod 143 = (7^{32} \bmod 143 \cdot 7^{32} \bmod 143) \bmod 143 = (16 \cdot 16) \bmod 143 = 256 \bmod 143 = 113$ Step 3: Use modular multiplication properties to combine the calculated mod C values $7^{106} \bmod 143$ $= (7^2 \cdot 7^8 \cdot 7^{32} \cdot 7^{64}) \bmod 143$ $= (7^2 \bmod 143 \cdot 7^8 \bmod 143 \cdot 7^{32} \bmod 143 \cdot 7^{64} \bmod 143) \bmod 143$ $= (49 \cdot 42 \cdot 16 \cdot 113) \bmod 143$ $= 3720864 \bmod 143$ $= 4$ $\therefore 7^{106} \bmod 143 = 4$ b) Determine all integers in the set $\{1, \dots, 11\}$ that are primitive roots of modulo 11 Ans: 2,6,7,8	
4.	a) Differentiate Initialization vector and Nonce in Block cipher modes operations. Explain the reason that you should use block cipher CBC mode instead of block cipher ECB mode. An IV is a nonce with an additional requirement: it must be selected in a nonpredictable way. That is, the IV can't be sequential; it must be random. CBC has the advantage over the ECB mode in that the XORing process hides plaintext patterns. Even if the first plaintext block and third plaintext block were the exact same segment of plaintext, it is highly unlikely that the first ciphertext block and third ciphertext block would be the same.	3 7

	b) Discuss primitive operations and key expansion procedure in IDEA and also Explain the steps involved in one round of IDEA to encrypt the plain text. • Each rounds makes use of 6 subkeys ($8 \times 6 = 48$) and the final output transformation uses four subkeys($48 + 4 = 52$ subkeys). • 128-bit key is divided into 8 parts K1 to K8 each of 16 bits. • Of which K1 to K6 is used for ROUND 1 operation. • For the remaining round subkeys are generated by performing key shifting. • The original key is shifted left circularly by 25 bits and hence 52 subkeys are generated. • Operations performed are Addition, Multiplication, XOR Each round involves a series of operation on the four data blocks using six keys. <table><tr><td>1. Multiply P1 and K1</td><td>i.e, $S1 = P1 \times K1$</td></tr><tr><td>2. Add P2 and K2</td><td>i.e, $S2 = P2 + K2$</td></tr><tr><td>3. Add P3 and K3</td><td>i.e, $S3 = P3 + K3$</td></tr><tr><td>4. Multiply P4 and K4</td><td>i.e, $S4 = P4 \times K4$</td></tr><tr><td>5. XOR step 1 and step 3</td><td>i.e, $S5 = S1 \oplus S3$</td></tr><tr><td>6. XOR step 2 and step 4</td><td>i.e, $S6 = S2 \oplus S4$</td></tr><tr><td>7. Multiply step 5 with K5</td><td>i.e, $S7 = S5 \times K5$</td></tr><tr><td>8. Add step 6 and step 7</td><td>i.e, $S8 = S6 + S7$</td></tr><tr><td>9. Multiply step 8 with K6</td><td>i.e, $S9 = S8 \times K6$</td></tr><tr><td>10. Add step 7 and step 9</td><td>i.e, $S10 = S7 + S9$</td></tr><tr><td>11. XOR step 1 and step 9</td><td>i.e, $S11 = S1 \oplus S9$</td></tr><tr><td>12. XOR step 3 and step 9</td><td>i.e, $S12 = S3 \oplus S9$</td></tr><tr><td>13. XOR step 2 and step 10</td><td>i.e, $S13 = S2 \oplus S10$</td></tr><tr><td>14. XOR step 4 and step 10</td><td>i.e, $S14 = S4 \oplus S10$</td></tr></table>	1. Multiply P1 and K1	i.e, $S1 = P1 \times K1$	2. Add P2 and K2	i.e, $S2 = P2 + K2$	3. Add P3 and K3	i.e, $S3 = P3 + K3$	4. Multiply P4 and K4	i.e, $S4 = P4 \times K4$	5. XOR step 1 and step 3	i.e, $S5 = S1 \oplus S3$	6. XOR step 2 and step 4	i.e, $S6 = S2 \oplus S4$	7. Multiply step 5 with K5	i.e, $S7 = S5 \times K5$	8. Add step 6 and step 7	i.e, $S8 = S6 + S7$	9. Multiply step 8 with K6	i.e, $S9 = S8 \times K6$	10. Add step 7 and step 9	i.e, $S10 = S7 + S9$	11. XOR step 1 and step 9	i.e, $S11 = S1 \oplus S9$	12. XOR step 3 and step 9	i.e, $S12 = S3 \oplus S9$	13. XOR step 2 and step 10	i.e, $S13 = S2 \oplus S10$	14. XOR step 4 and step 10	i.e, $S14 = S4 \oplus S10$	
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5.	A person wants to share a plaintext “123456ABCD132536” to his friend in the opposite side through social network. He/she uses Data Encryption Standard (DES) algorithm for session encryption during his communication and assume that he/she uses, the key as “AABB 0918 2736 CCDD”. Show the key generated for the first two rounds. Parity Drop Table	10																												

57	49	41	33	25	17	09	01
58	50	42	34	26	18	10	02
59	51	43	35	27	19	11	03
60	52	44	36	63	55	47	39
31	23	15	07	62	54	46	38
30	22	14	06	61	53	45	37
29	21	13	05	28	20	12	04

Compression Box Table

14	17	11	24	01	05	03	28
15	06	21	10	23	19	12	04
26	08	16	07	27	20	13	02
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

Key in Hexa to Binary

AABB 0918 2736 CCDD

1010 1010 1011 1011 0000 1001 0001 1000 0010 0111 0011 0110 1100 1100
1101 1101

Key 1: 000110010100110011010000011100101101111010001100

Key 2: 010001010110100001011000000110101011110011001110
